

Empirical Proof Record: REFUTED

2026-05-23T23:11:36.989637+00:00

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Proof ID: a250815fd7fb7735d1b8664987d13033...

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1. Claim

On real technical-domain text (Linux kernel commit messages) that clears Kimera's GWF, the substrate's walker reaches sustained traversal (`halt_mode == 'exhausted'`) in $\geq 60\%$ of cycles — i.e. the topology layer engages on technical reasoning instead of collapsing to `amplitude_death`.

- **Operationalisation:** fraction of GWF-cleared Linux-kernel-commit cycles for which `result.halt_mode == 'exhausted'`
- **Threshold:** `sustained_traversal_rate_on_cleared` ≥ 0.6 fraction
- **H0:** $P(\text{halt}=\text{'exhausted'} \mid \text{gwf cleared}) < 0.6$
- **H1:** $P(\text{halt}=\text{'exhausted'} \mid \text{gwf cleared}) \geq 0.6$

2. Verdict

REFUTED — observed 0.125

walker reached sustained traversal in 4/32 GWF-cleared cycles (12.5%); `amplitude_death` in 28/32 (87.5%); GWF blocked 8/40 (20.0%); halt modes observed: {'`amplitude_death`': 28, '`exhausted`': 4}; dissonance median=14 (range 0-35); Φ median=0.632; 40 cycles, 0 adapter errors

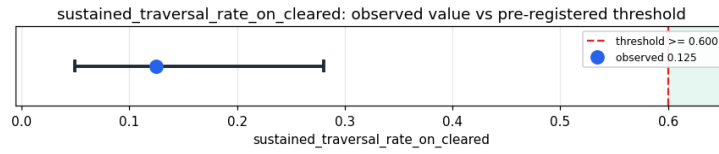


Figure 1: sustained_traversal_rate_on_cleared – observed value with Wilson 95% CI.

3. Evidence

Pillar	Statistic	Value
0.topology.sustained_traversal	sustained_traversal_rate_on_cleared	0.125
0.topology.amplitude_death_rate	amplitude_death_rate_on_cleared	0.875
0.topology.halt_mode_distribution	halt_modes_observed_count	2.0
0.topology.gwf_block_rate	gwf_block_rate_on_linux	0.2
0.topology.dissonance_intensity	dissonance_events_count_median	14.5
0.topology.phi_distribution	phi_value_median	0.6323452260010114

4. Pre-registration

- Registered at: 2026-05-23T23:09:41.097334+00:00
- Config hash: 8828d82eff83d5002e25687d42f2897a...
- Data hash: 92a34cc4275008279006eeec2c19cd2b...

5. Signature

2aabd94e3460bfc28f5a2e65485ae7a34260ab084dd7b794c4fc75712259a87d